

Simulation and Modeling

Lecture 04

Simulation Development Example Single Server Queueing System (SSQS)

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Computational Model

Computation Model (2/7)

Conceptual Model

- State Variables
- Set of Events
- Input Variables
- Output Variables

Specification Model

- State Equations
- Output Equations
- Input Data Modeling

Computational Model

System

- State Variables
- Events
 - Event Handling Alg
- Input variables
 - Random generator
- Output Variables
 - Output Var update Alg

Simulator

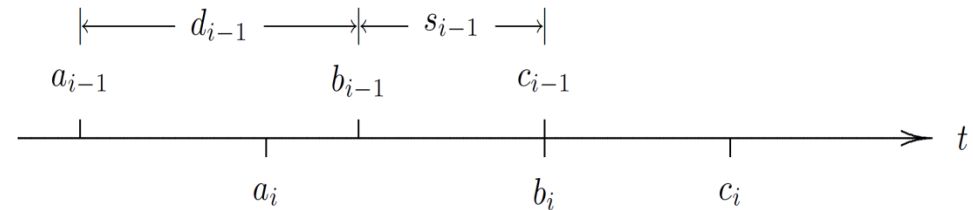
- Clock
 - Update mechanism
- EventList
- Scheduler
 - Run ()
 - Trigger Events
 - Call the handler

Simple Computation Model (1/2)

Delay in queue (2/2)

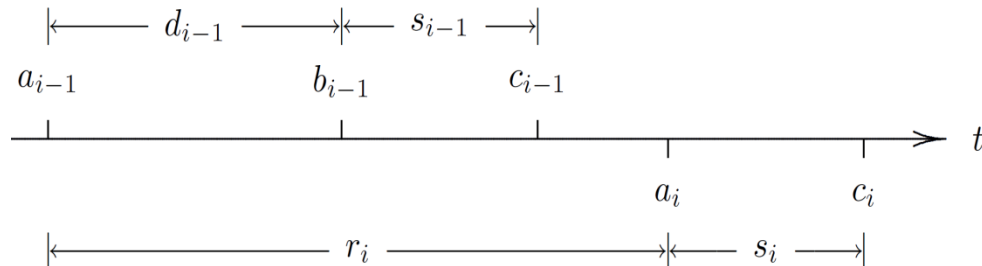
- Case I: $a_i < c_{i-1}$, job i arrives before job $i - 1$ departs

$$d_i = c_{i-1} - a_i$$



- Case II: $a_i \geq c_{i-1}$, job i arrives after job $i - 1$ departs

$$d_i = 0$$



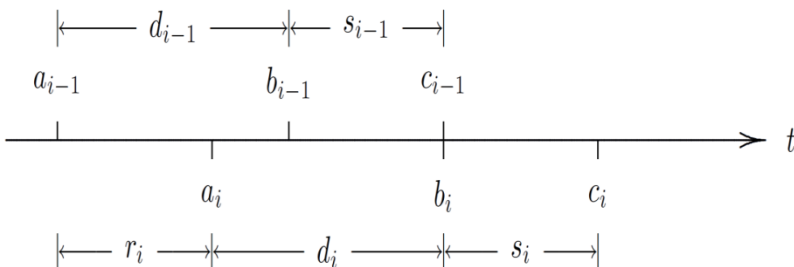
Simple Computation Model (2/2)

4. Output Equations (5/5)

```

c0 = 0.0;
i = 0;
while ( more jobs to process ) {
    i++;
    ai = GetArrival();
    if (ai < ci-1)
        di = ci-1 - ai;
    else
        di = 0.0;
    si = GetService();
    ci = ai + di + si;
}
n = i;
return d1, d2, ..., dn;

```



- For the last job, completion time is

$$c_{10} = 320 + 26 + 30 = 376$$

i	:	1	2	3	4	5	6	7	8	9	10
a_i	:	15	47	71	111	123	152	166	226	310	320
d_i	:	0	11	23	17	35	44	70	41	0	26
s_i	:	43	36	34	30	38	40	31	29	36	30

Computational Model: Simulator

Member variables

- Clock_
- eventList_

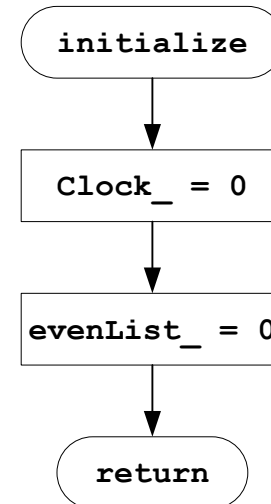
Member Functions

- Initialize ()
- Run ()
- activate (Event *e)
- Event* deactivate ()
- Event* removeEvent ()

Computational Model: Simulator

initialize () function

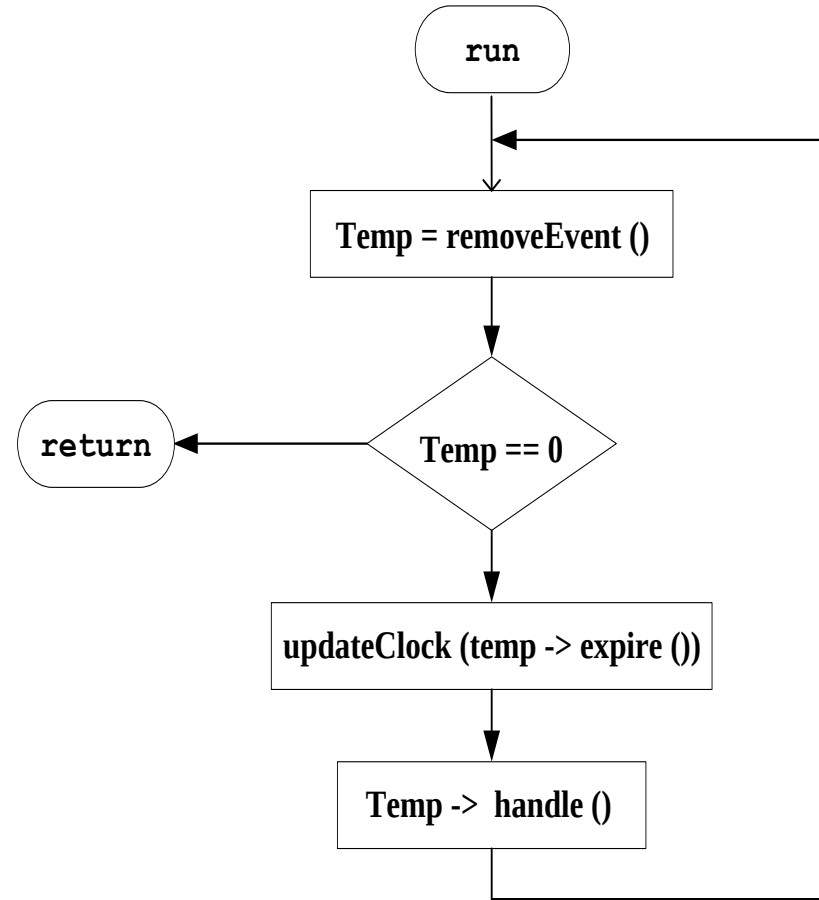
- Initialize the simulator
- Set the clock_ to 0
- Set the eventList_ to null



Computational Model: Simulator

run () function

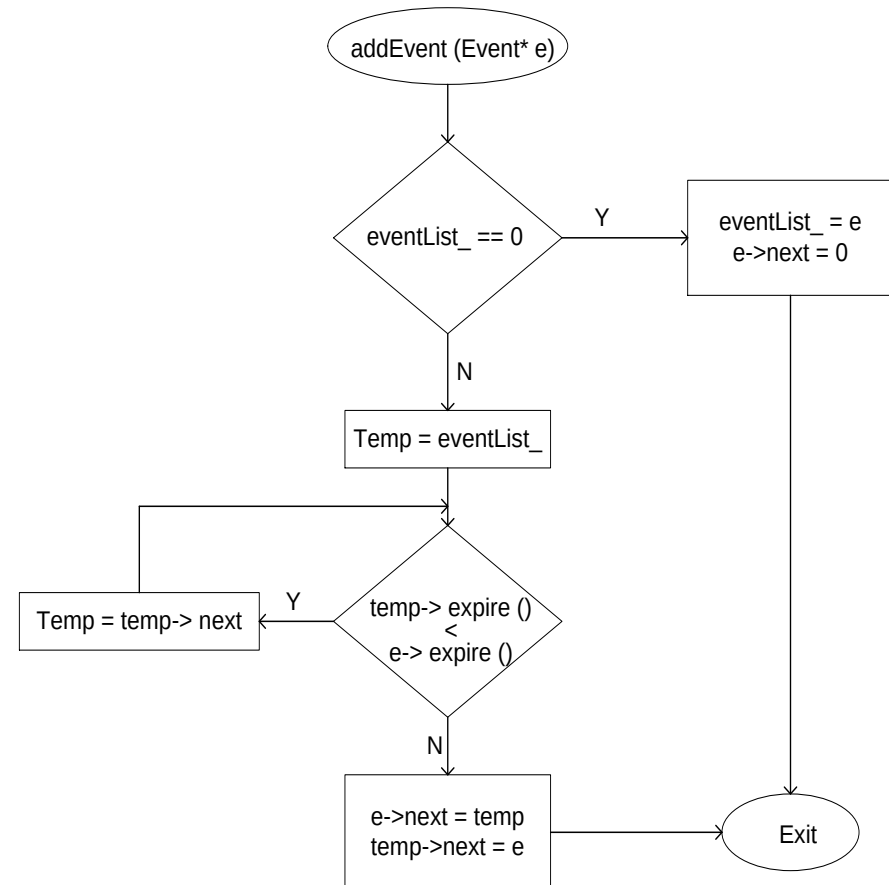
- Schedule the events
 - First event in the eventList_ is scheduled
 - If there is no event in eventList_, simulation is terminated
- Update the simulation time clock_ to the current time
- Call the event handling function
- Move to the next event in the list



Computational Model: Simulator

activate() function

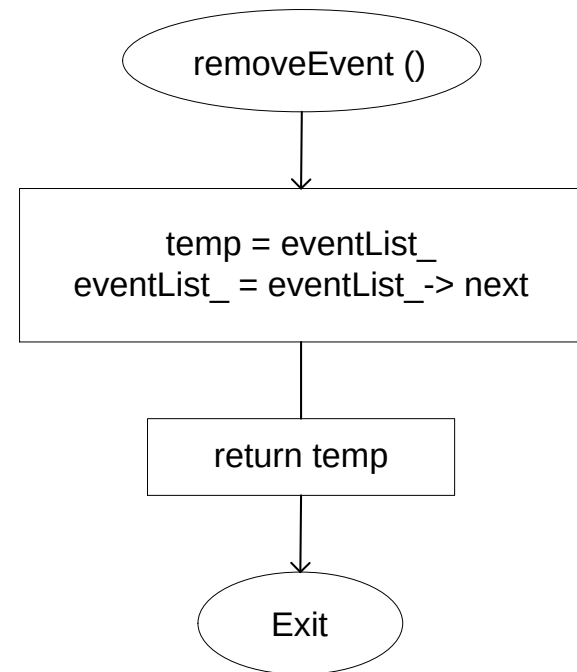
- Add an event in the eventList_
- Check whether the event is added
 - in an empty list
 - as first element in the list
 - in the middle of the list
 - as last element of the list
- Flow chart shows two cases
 - Empty list
 - Middle of a list



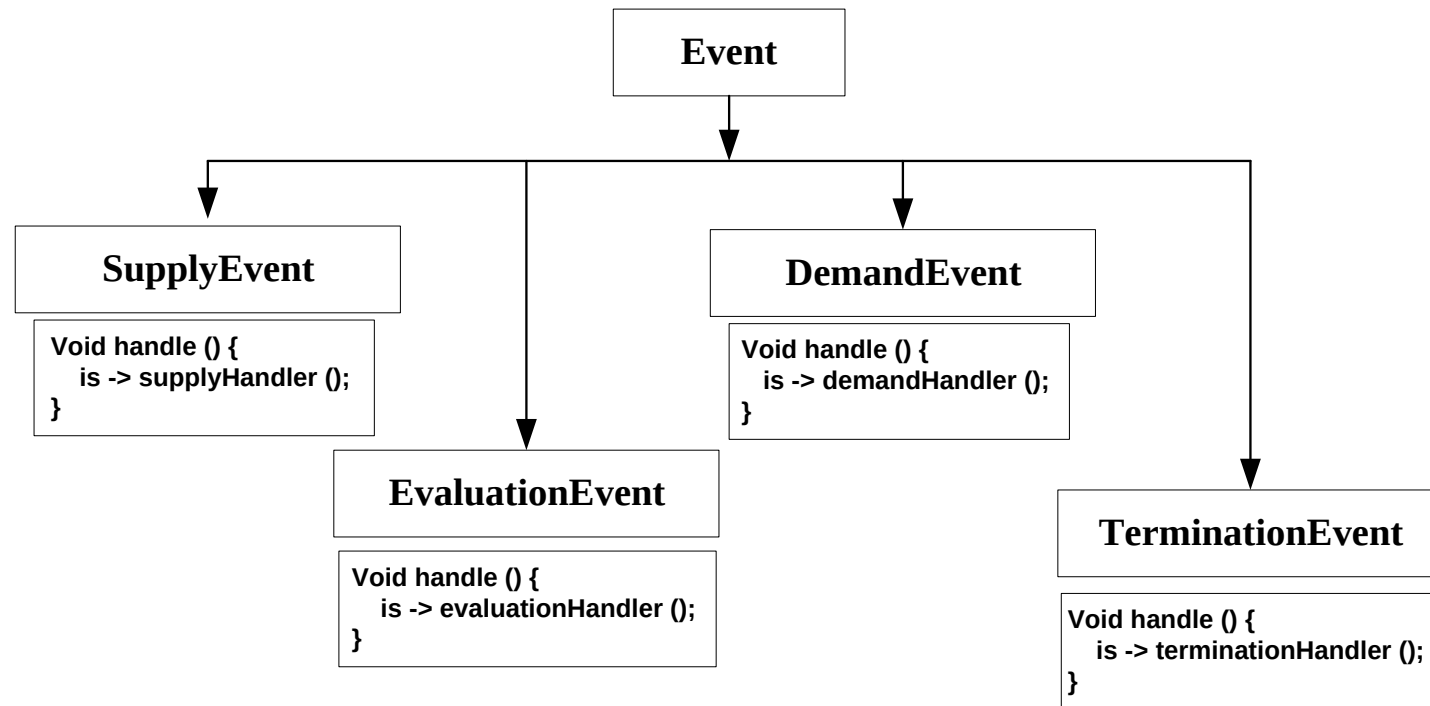
Computational Model: Simulator

removeEvent () function

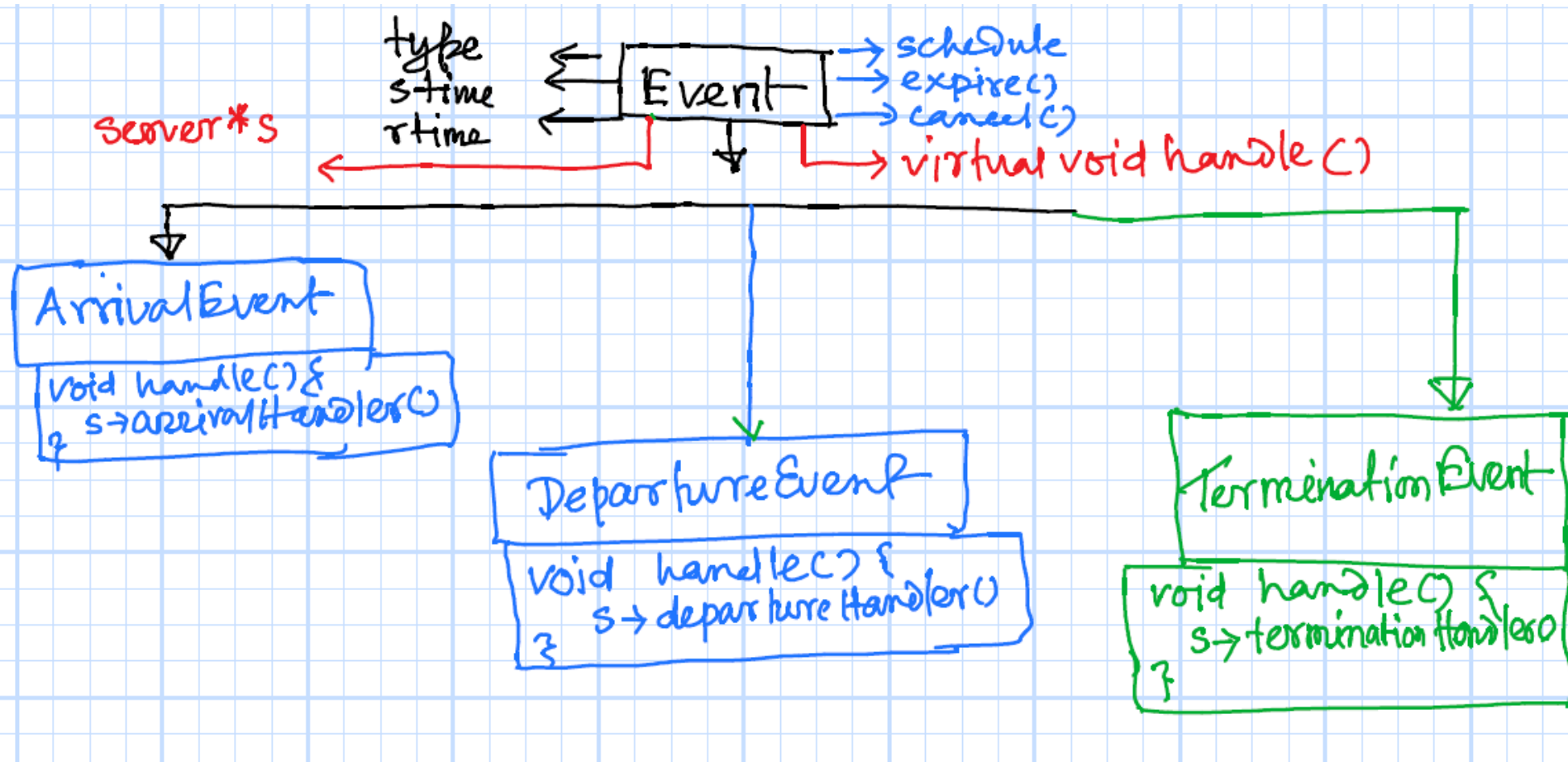
- Remove the first event of the eventList_
 - Return the event
- If the list is empty returns null



Computational Model: Events



Computational Model: Events



Computational Model: Server

State variables

- status
- length
- Job
 - id
 - arrivalTime
 - delay
 - serviceTime
 - interArrivalTime
- Special Variables
 - lastEventTime
 - lastArrivalTime

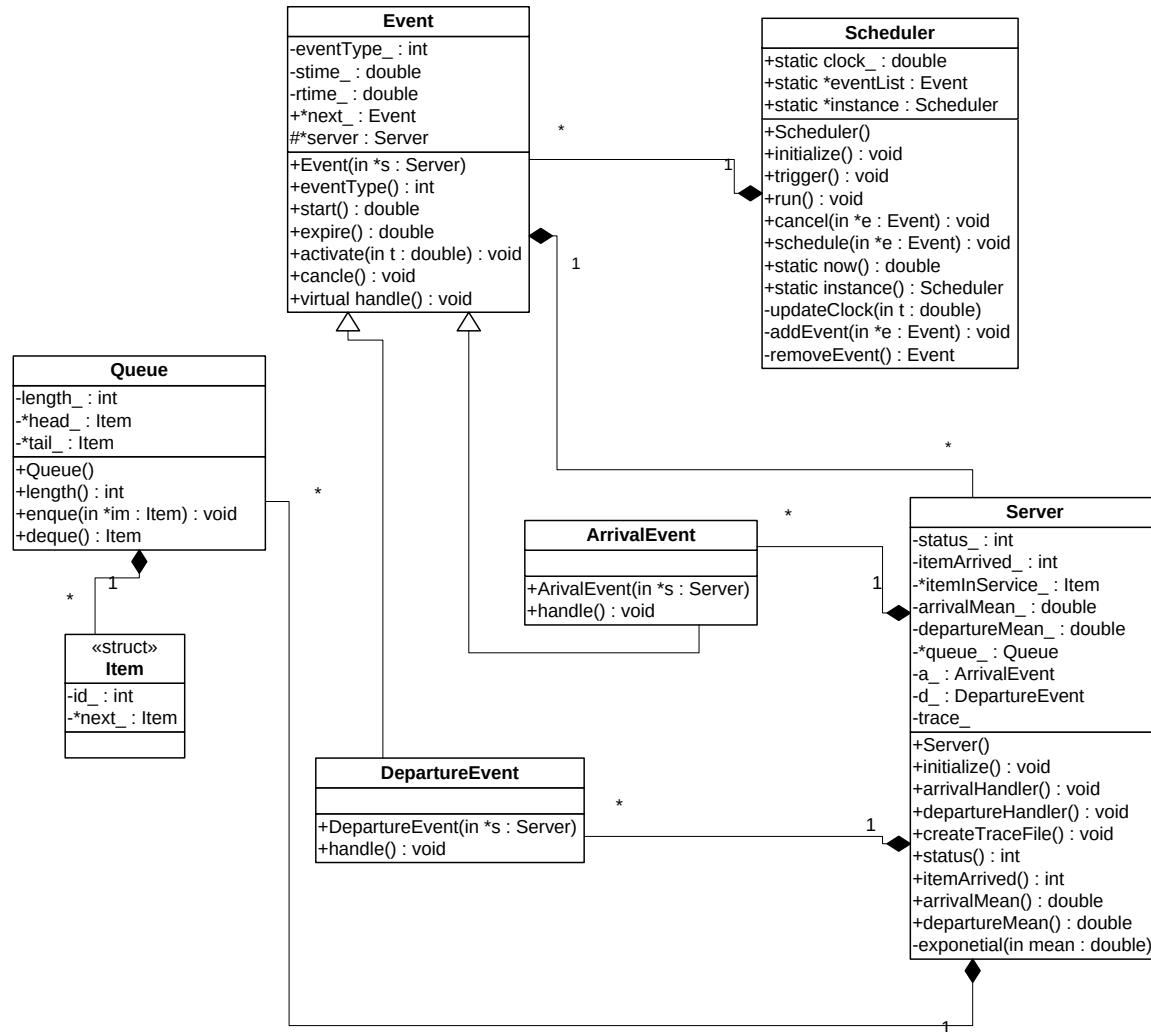
Statistical Variables

- Job Average
 - totalDealy
 - totalInterarrivalTime
 - totalServiceTime
- Time Average
 - totalJobTimeInQ
 - totalJobTimeInServer

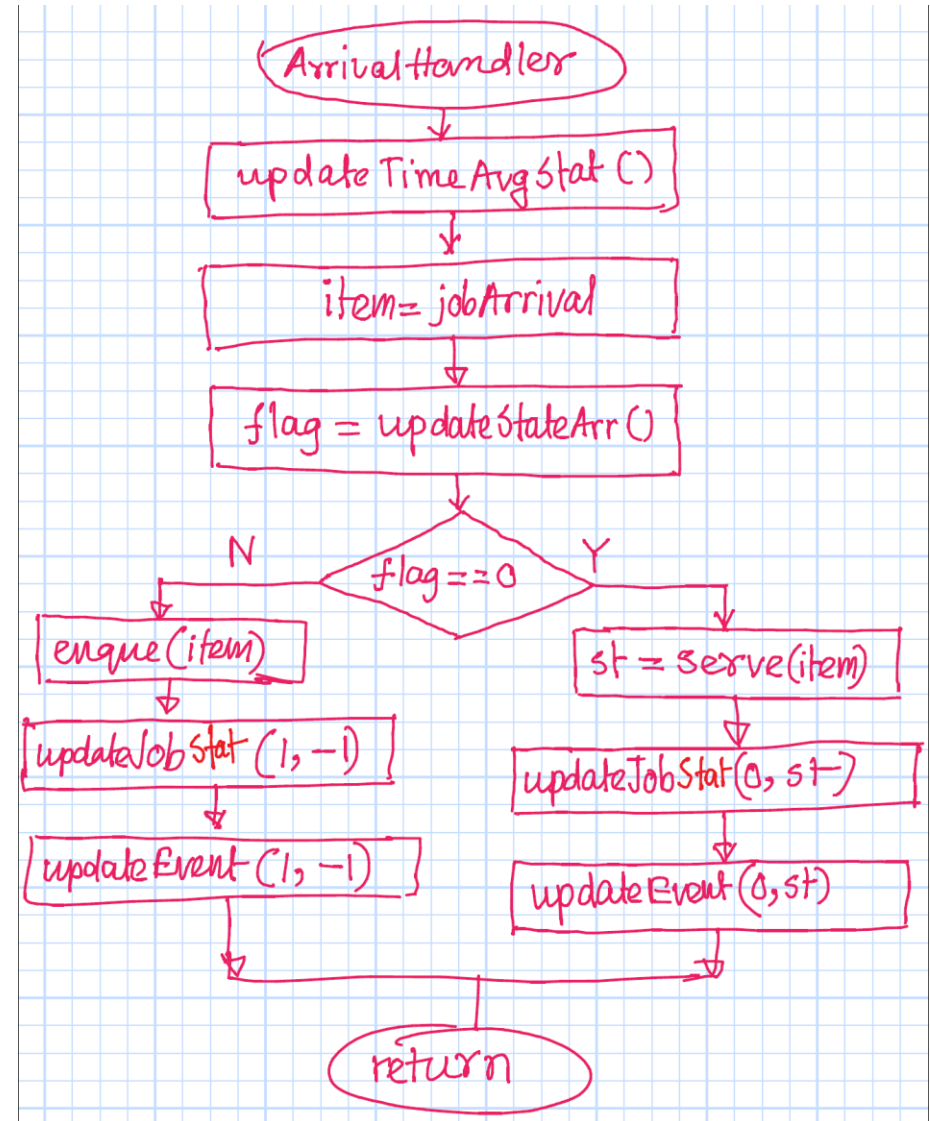
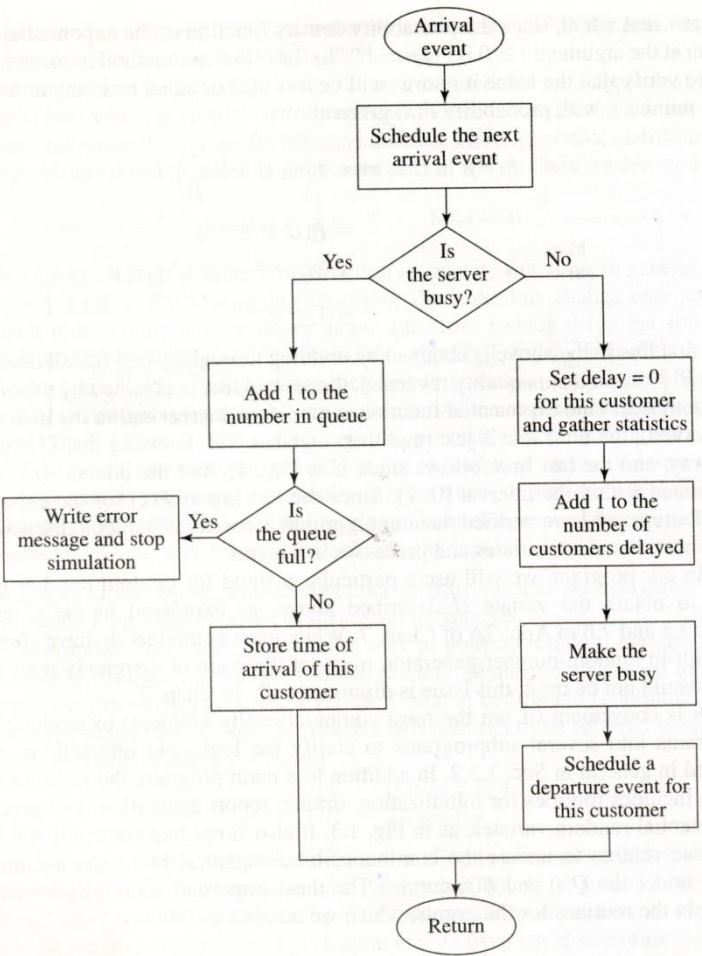
Member Functions

- Initialize ()
- arrivalHandler ()
- departureHandler () ()

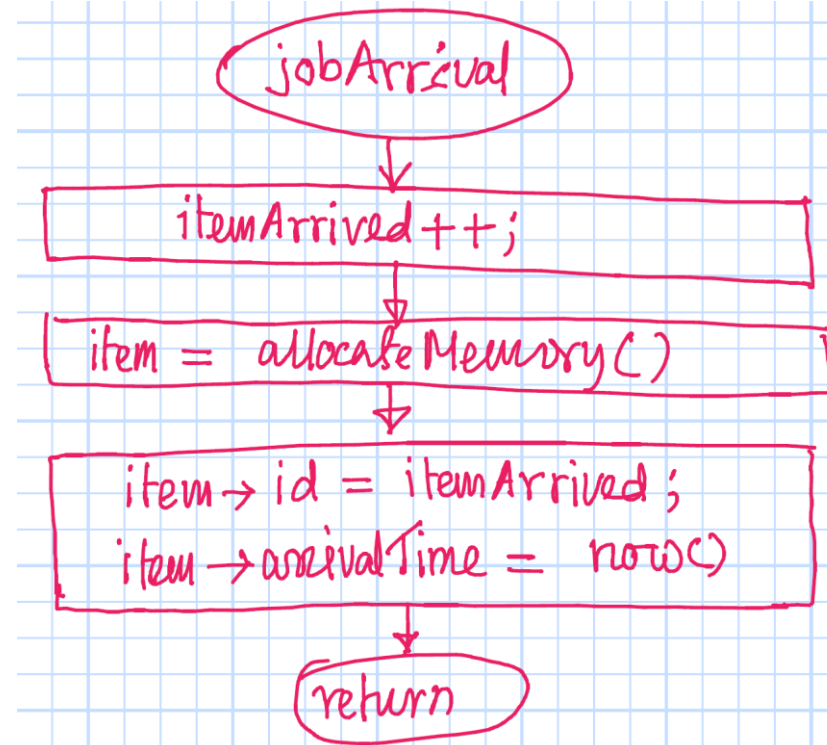
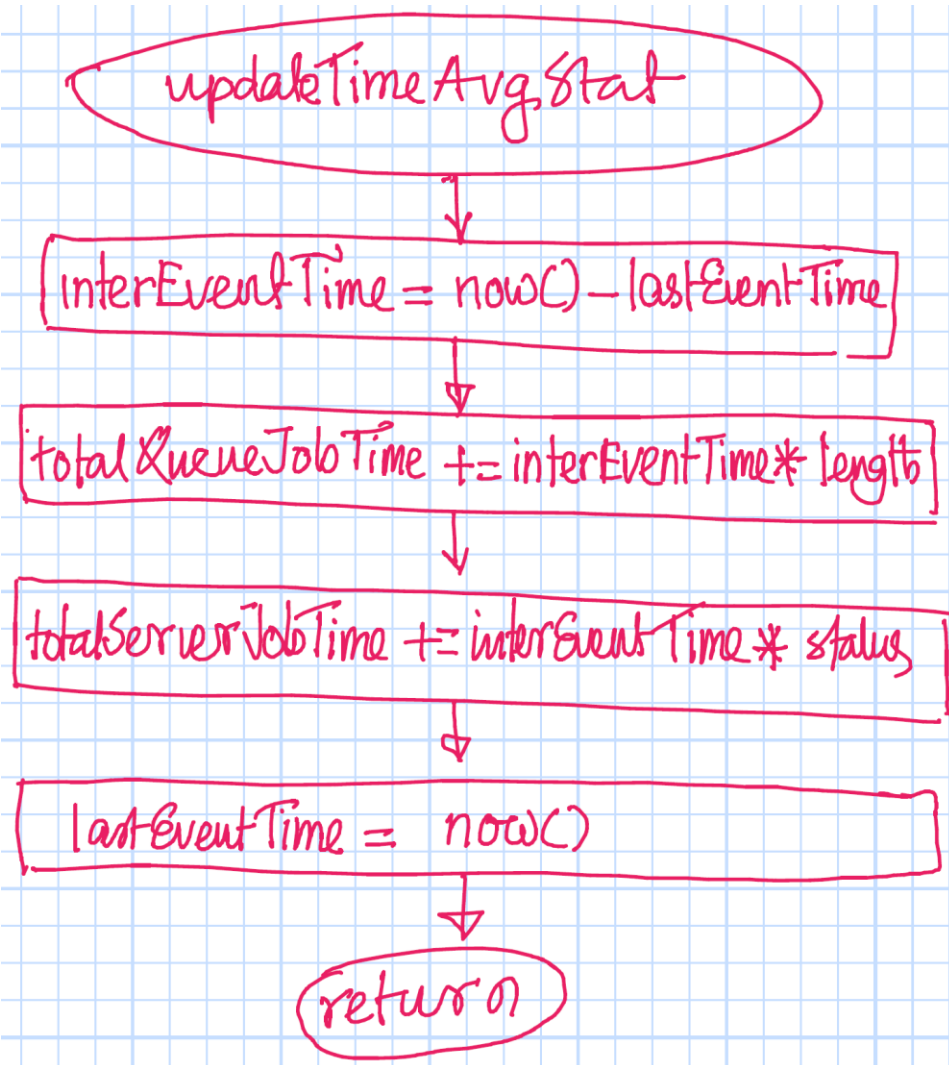
SSQS – Step 4: Computation Model



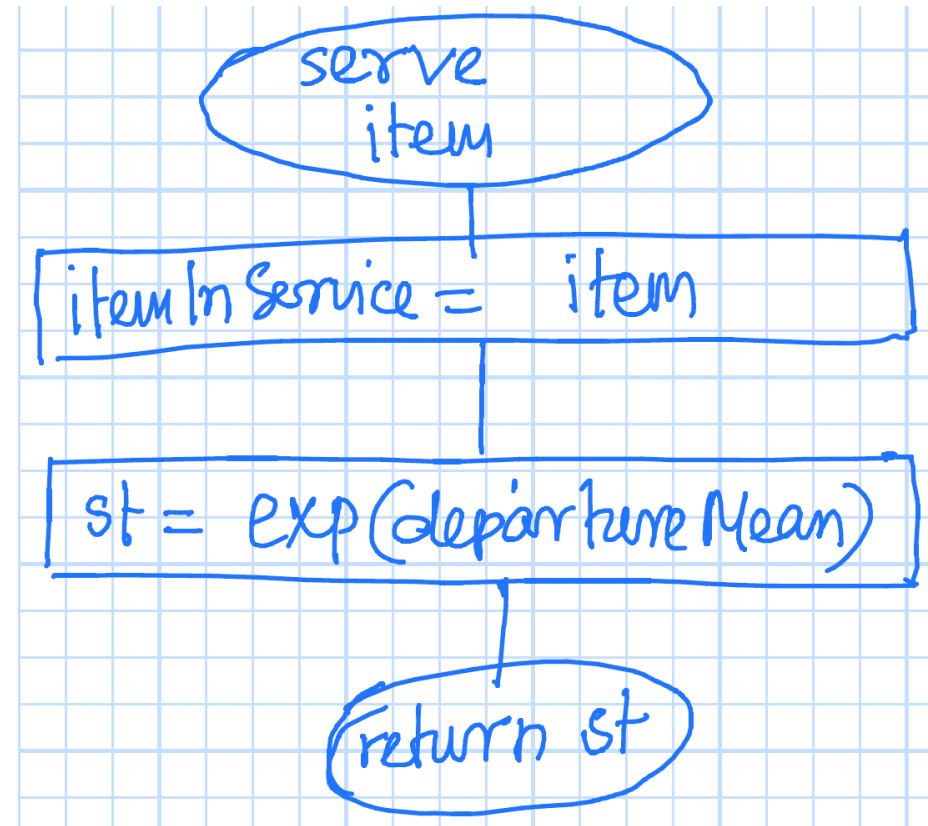
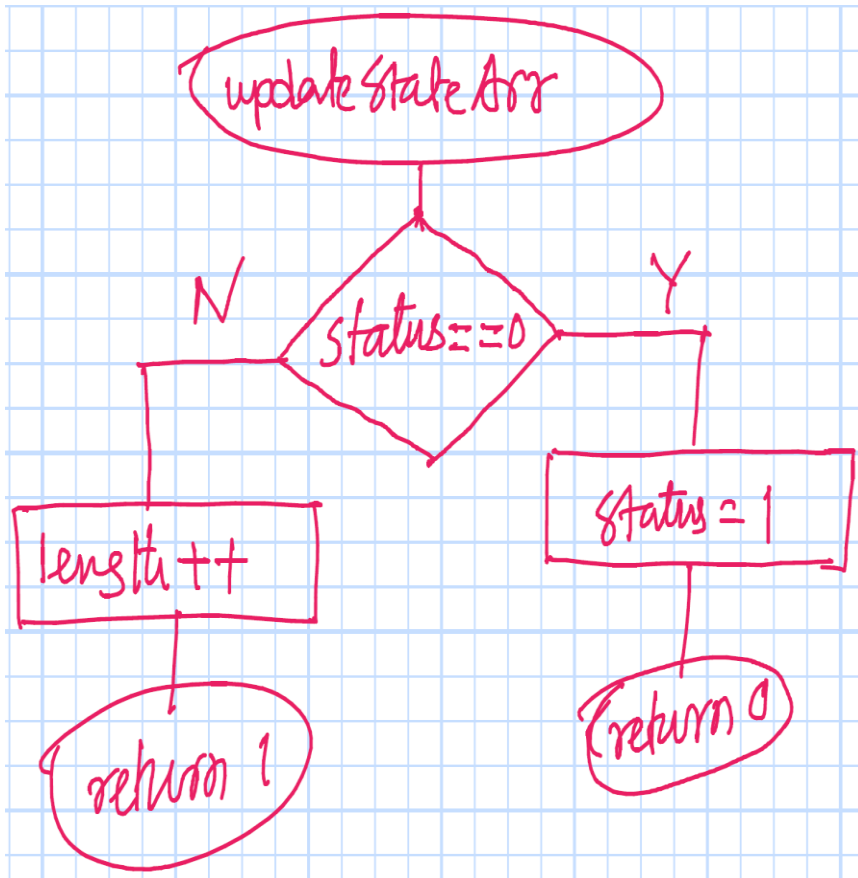
SSQS – Step 4: Computation Model



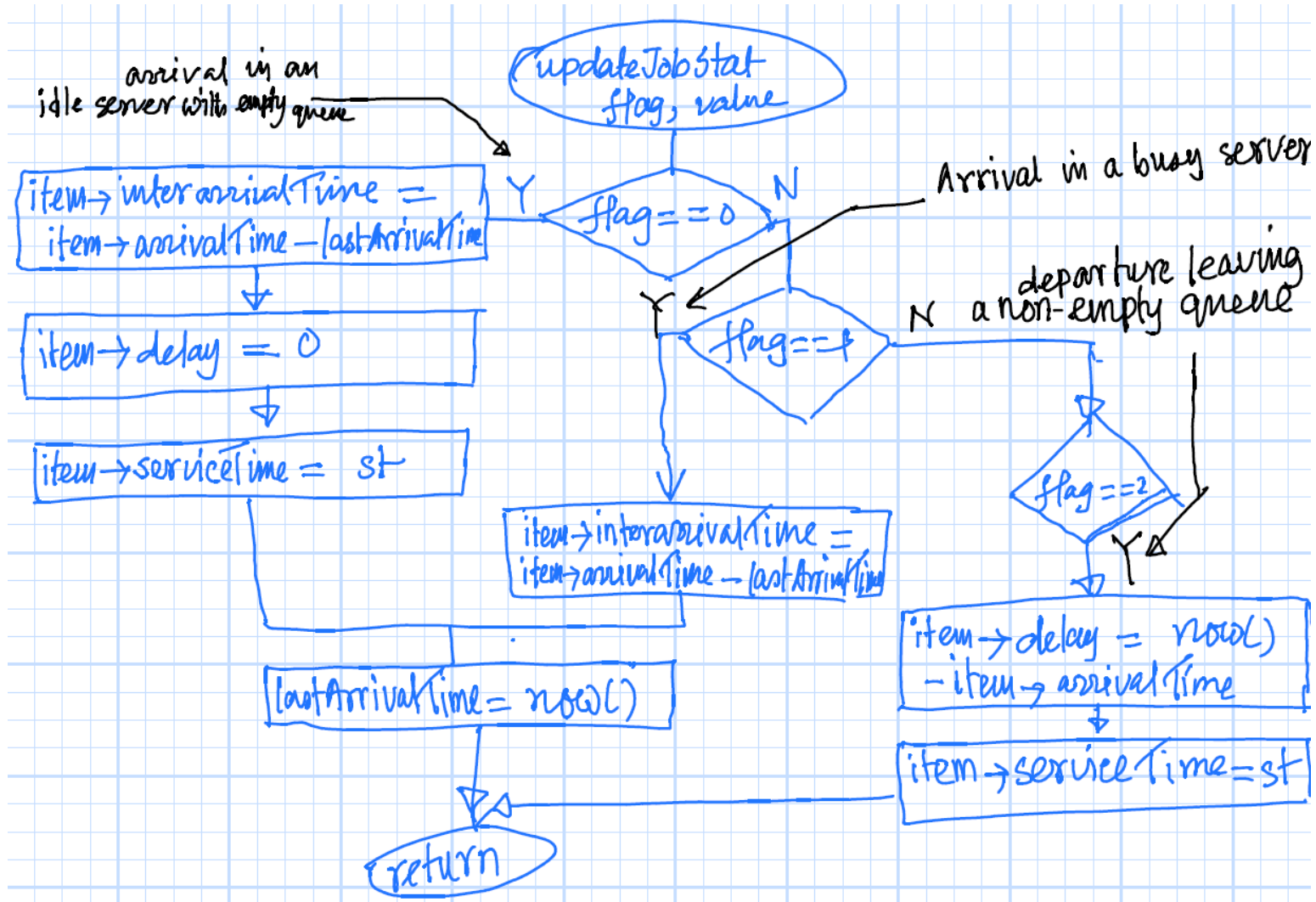
SSQS – Step 4: Computation Model



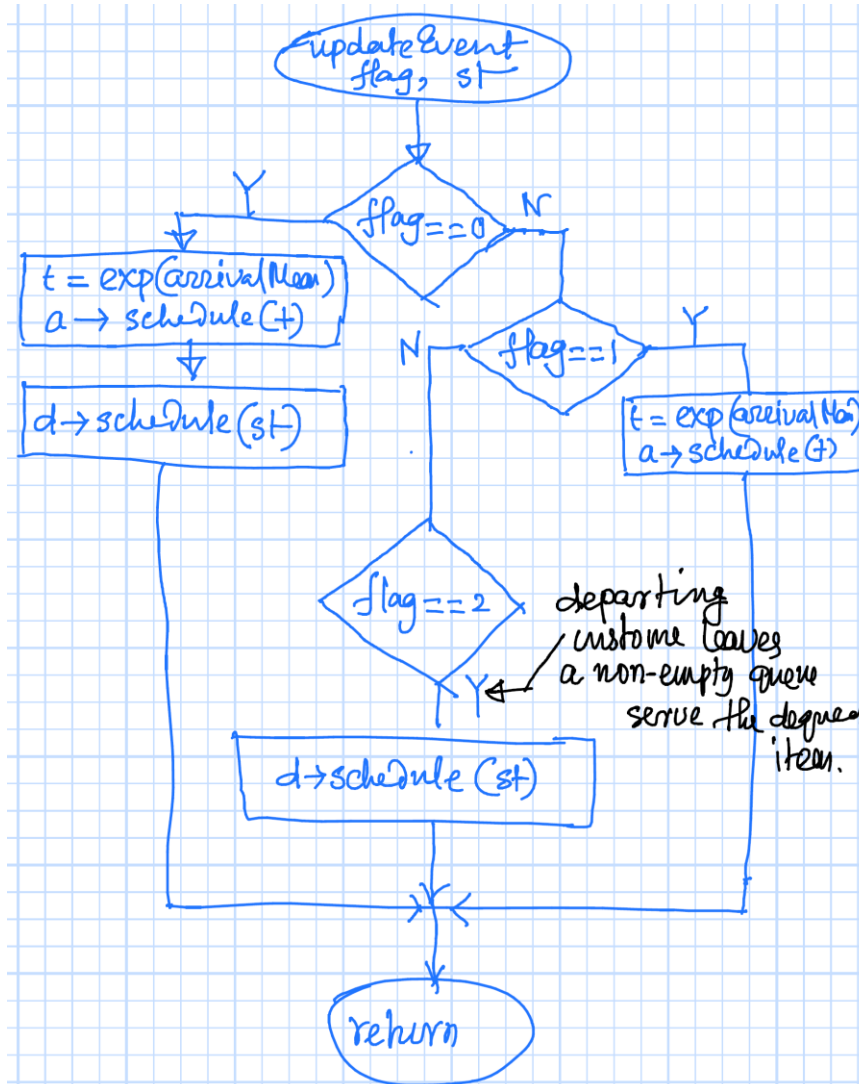
SSQS – Step 4: Computation Model



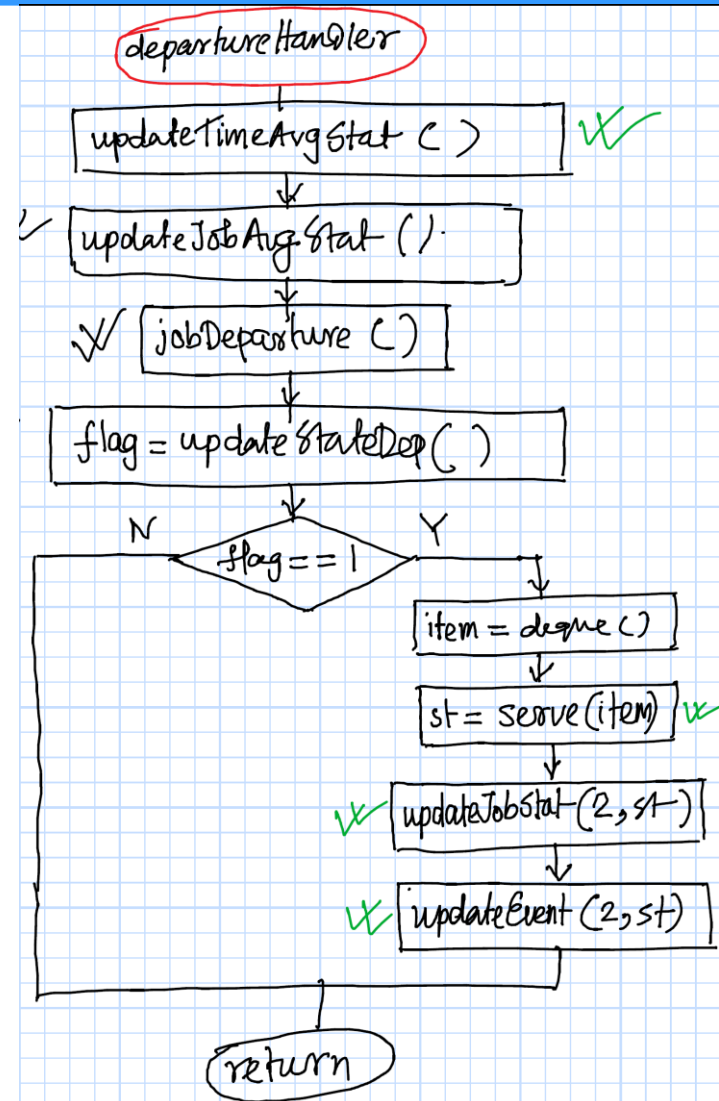
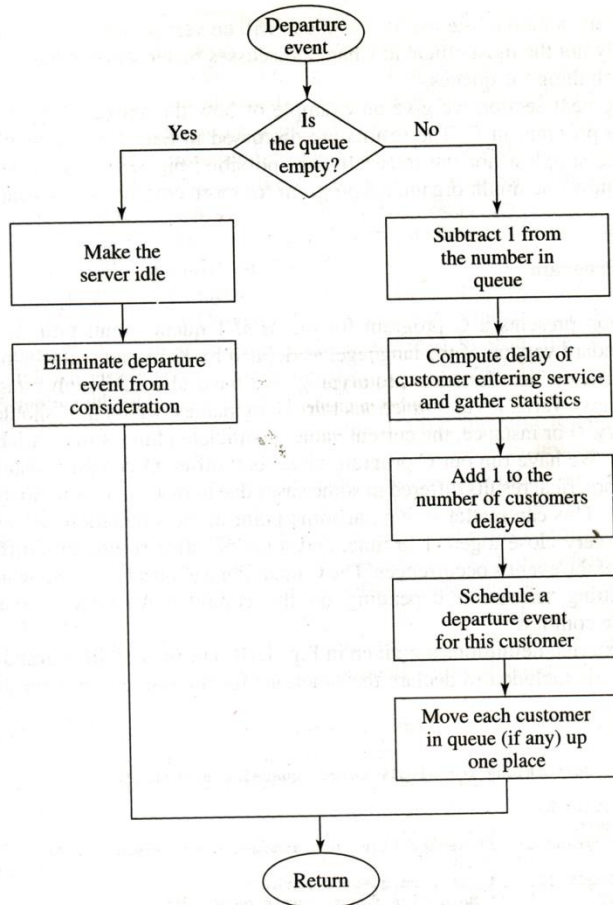
SSQS – Step 4: Computation Model



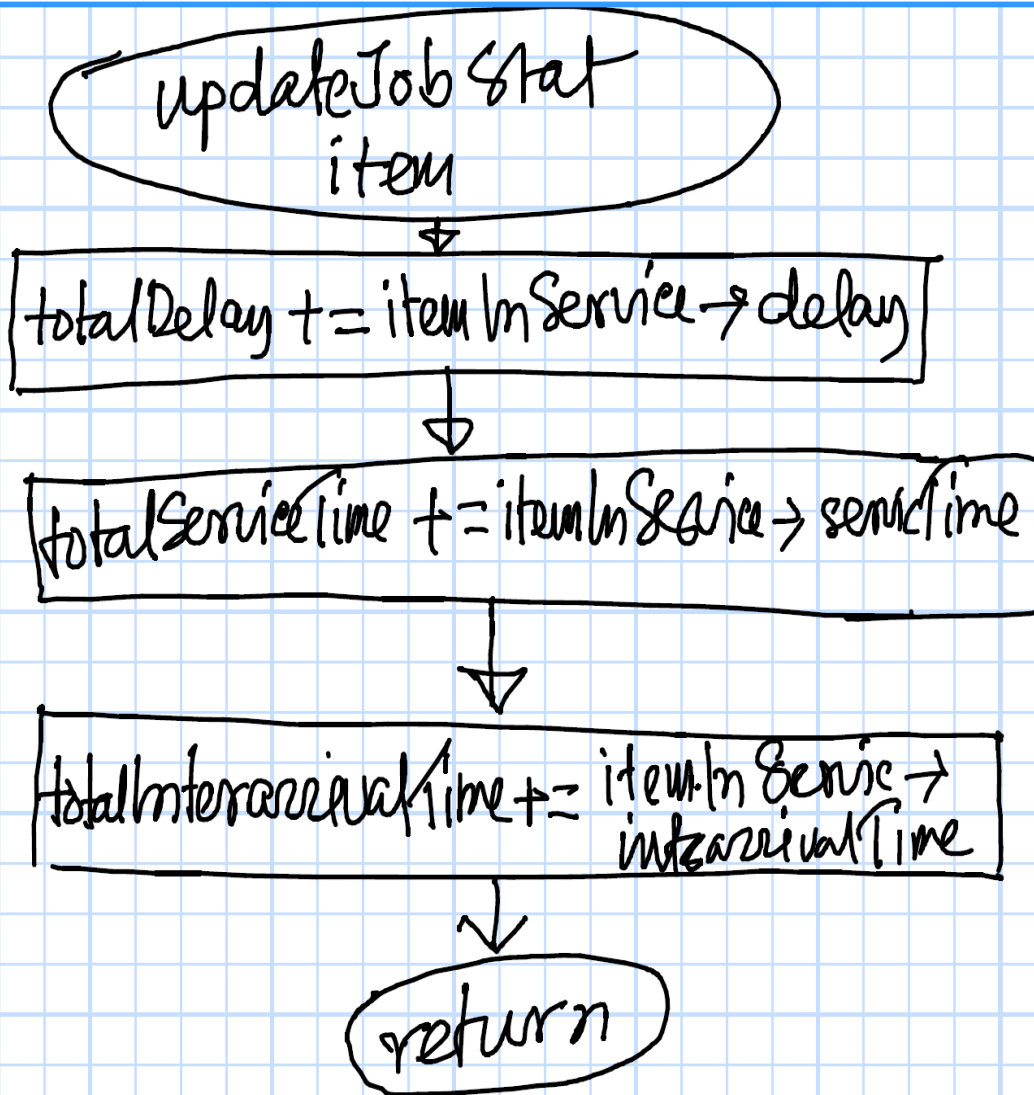
SSQS – Step 4: Computation Model



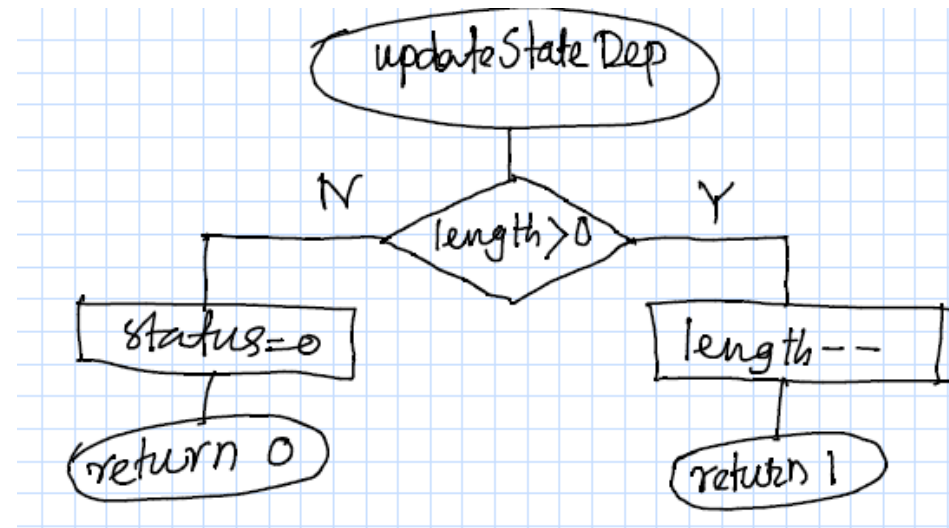
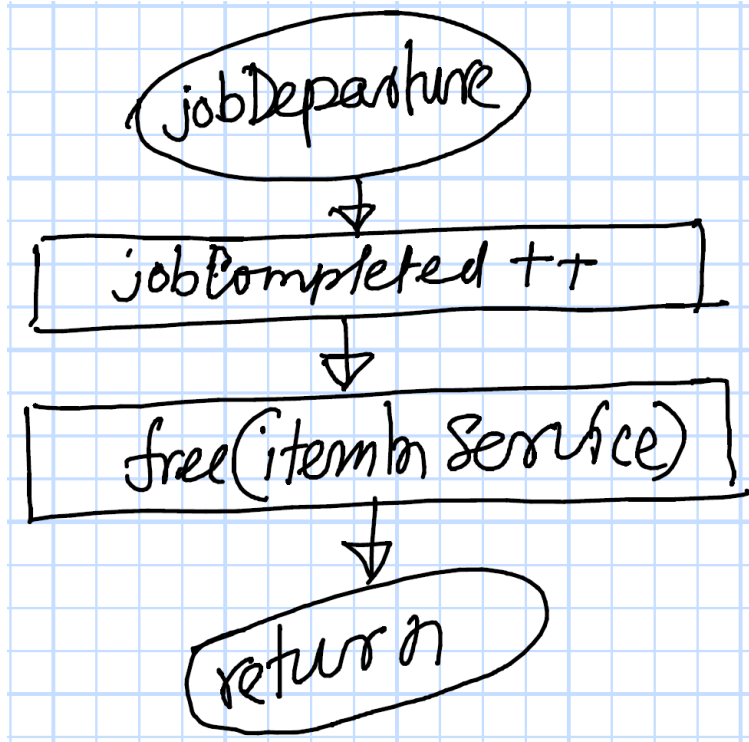
SSQS – Step 4: Computation Model



SSQS – Step 4: Computation Model



SSQS – Step 4: Computation Model



-
- | • Interarrival time | --- Service Time |
|---------------------|------------------|
| • 0.4 (0.4) | 2.0 (2.4), |
| • 1.2 (1.6) | 0.7 (3.1), |
| • 0.5 (2.1) | 0.2 (3.3), |
| • 1.7 (3.8) | 1.6 (4.9), |
| • 0.2 (4.0) | 3.7 (8.6), |
| • 1.6 (5.6) | 0.6 (9.2), |
| • 0.2, (5.8) | |
| • 1.4, (7.2) | |
| • 1.9, (9.1) | |

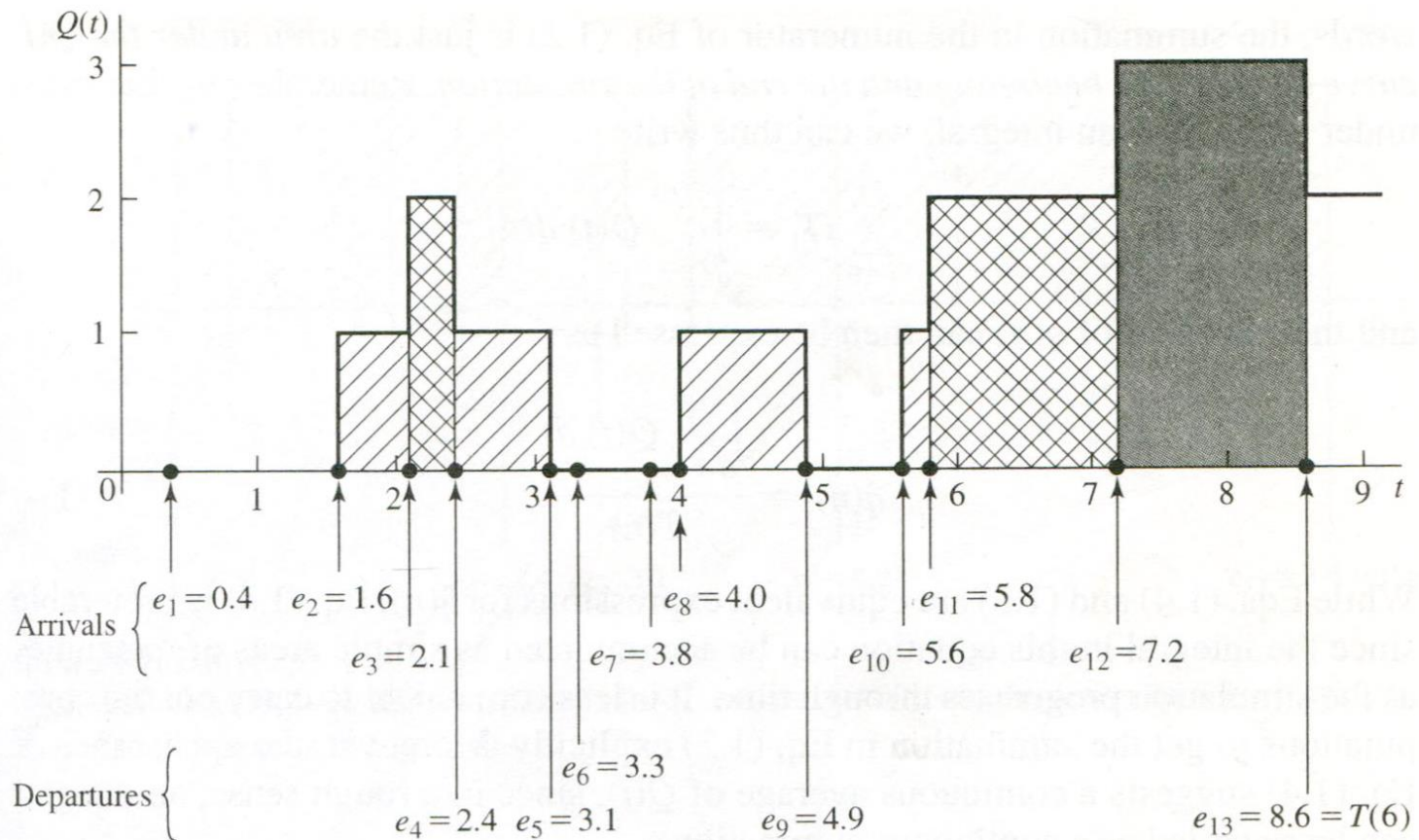


FIGURE 1.5

$Q(t)$, arrival times, and departure times for a realization of a single-server queueing system.

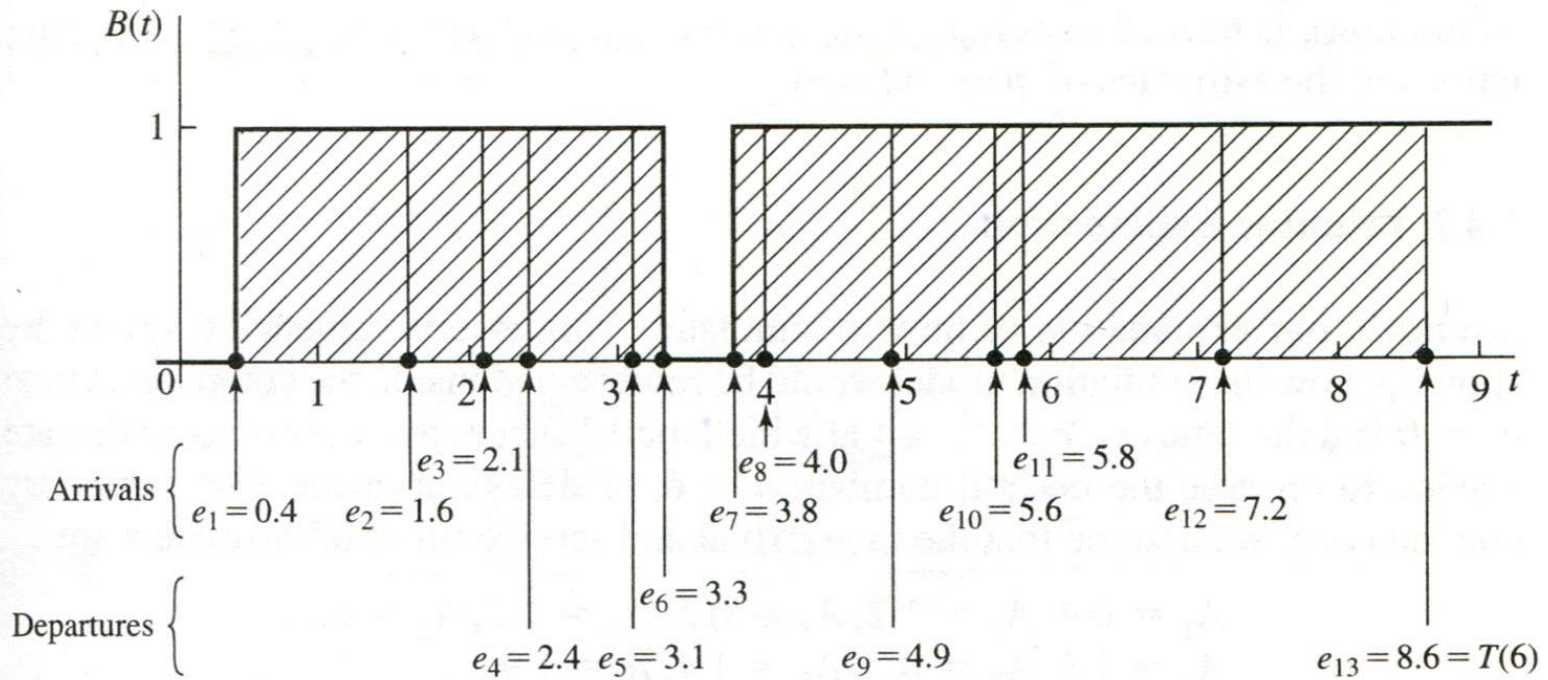
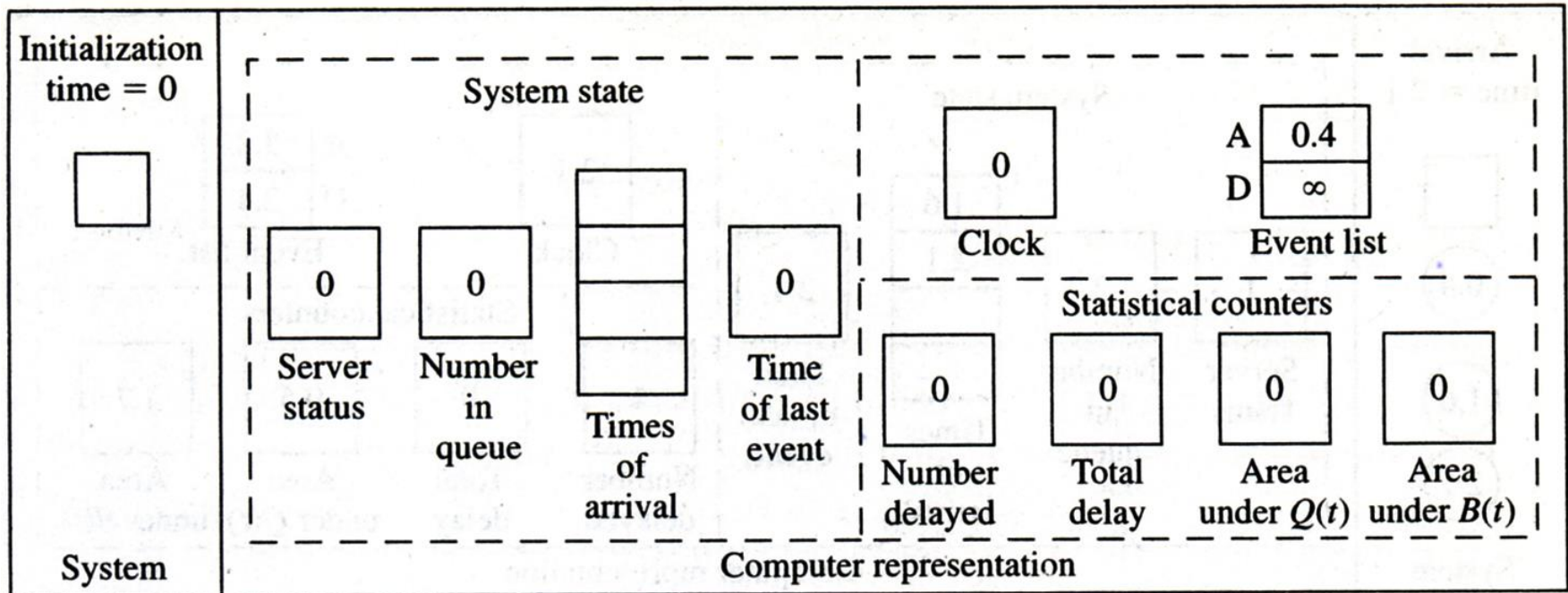


FIGURE 1.6

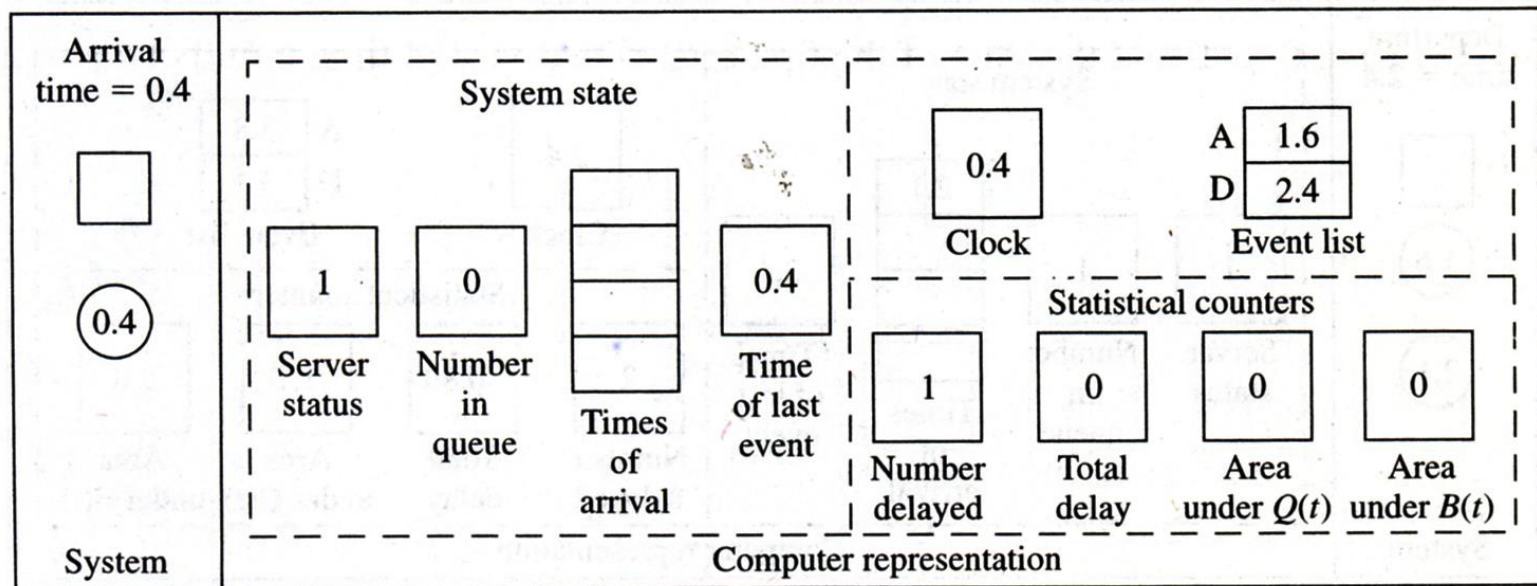
$B(t)$, arrival times, and departure times for a realization of a single-server queueing system (same realization as in Fig. 1.5).

SSQS – Step 4: Computation Model



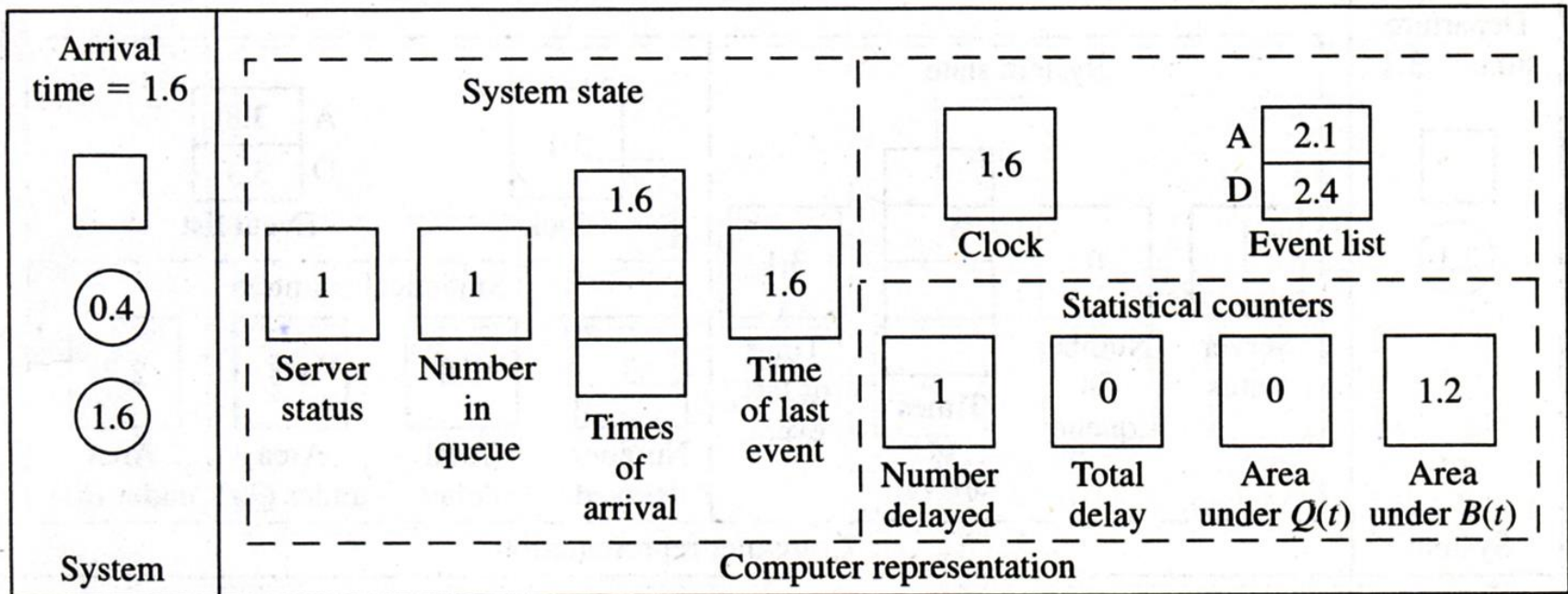
(a)

SSQS – Step 4: Computation Model



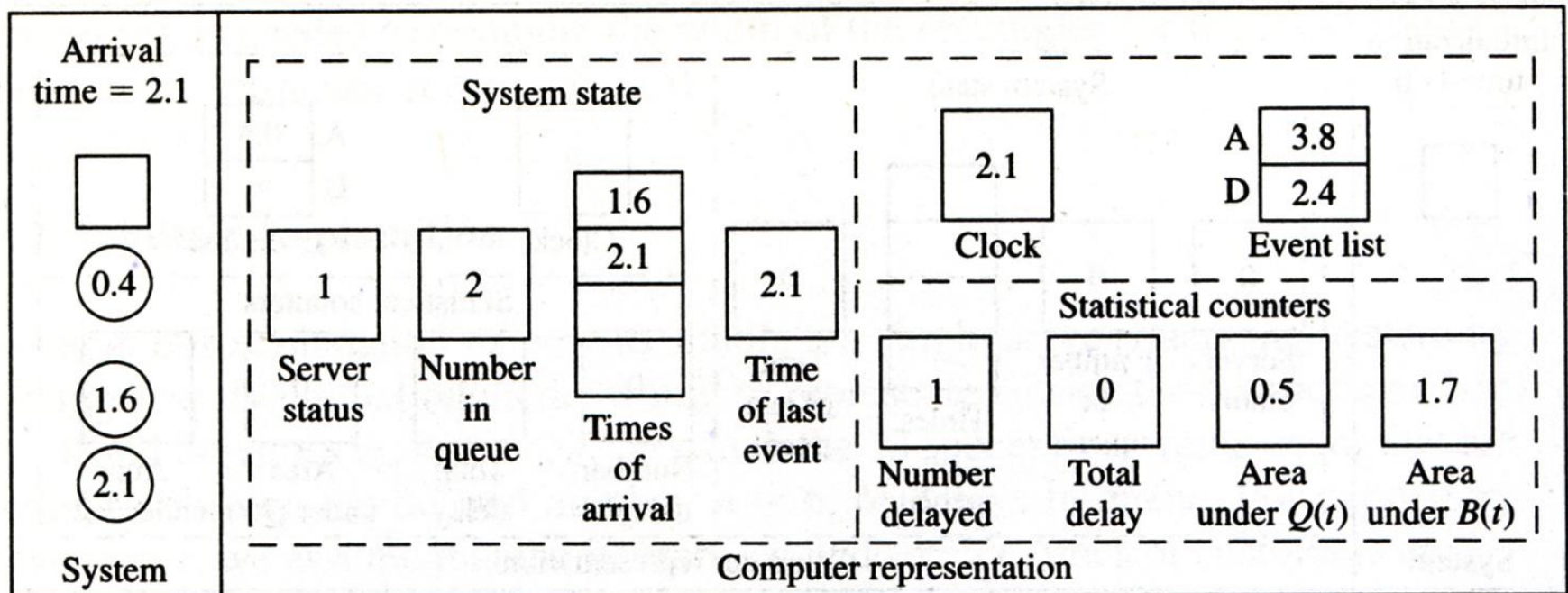
(b)

SSQS – Step 4: Computation Model



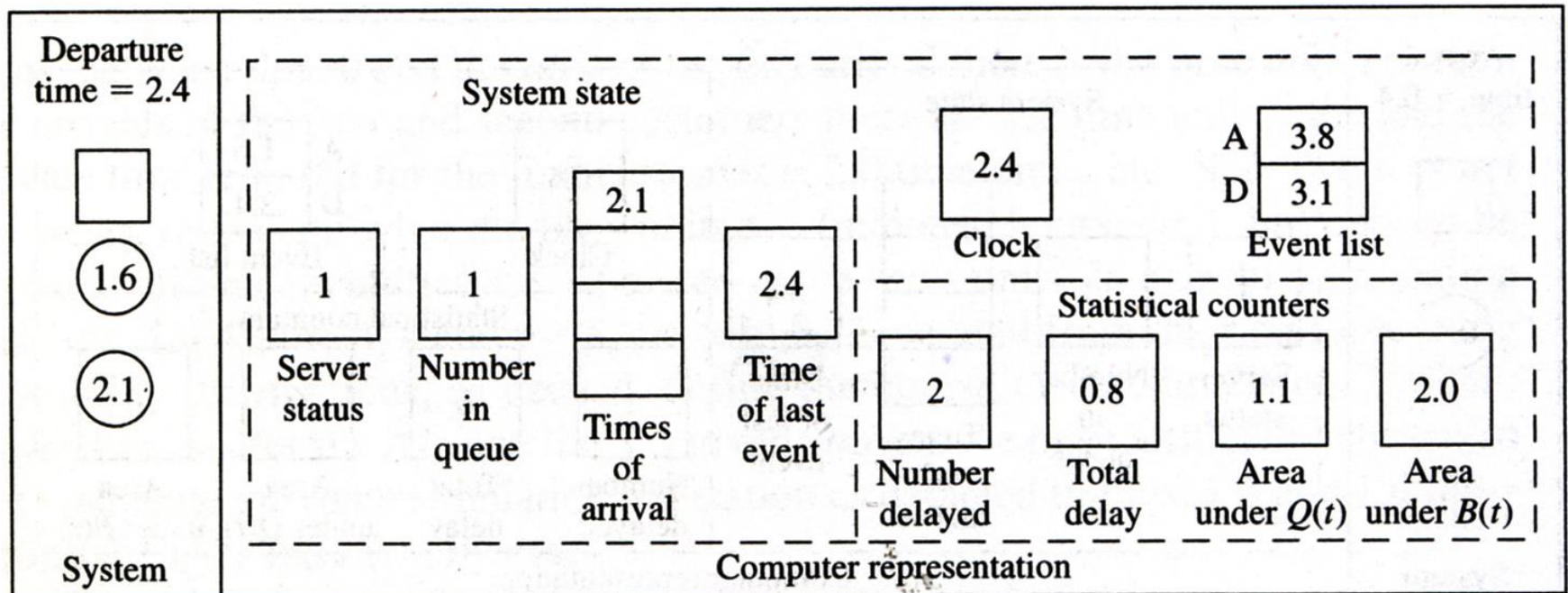
(c)

SSQS – Step 4: Computation Model



(d)

SSQS – Step 4: Computation Model



(e)